

CS 6762
Fall 2023
SP, ML and FC
Homework 4

Answer all questions. Work alone. Total 100 points. Type or write neatly.

1. What are the SASO properties? (5 points)
2. What are the 5 steps required to perform system ID and name two metrics that can be used for assessing quality of model fit to the data? (10 points)
3. Write general difference equations for a 2nd order 3-input, 2-output MIMO system. (5 points)
4. What is the z-transform for each of the following signals? (10 points)
 - a. $U(k) = (1, 5, 10, 0, 20, 0, \dots)$
 - b. $U(k) = (1, 1, 1, 1, 1, \dots)$
5. What is the inverse z-transform for the following: $U(z) = z/(z-1) + z/(z-.3)$. (10 points)
6. Given the following z transform what is the output steady state, $y(ss)$? $Y(z) = -.5z/((z-1)(z-.4))$ (5 points)
7. If you have a system with the following transfer function $G(z) = \frac{6 - 5z^{-1} + 10z^{-2} + 6z^{-3}}{1}$ what is the time domain output of this system to a unit impulse input? (10 points)
8. If I have a system where input $u(k)$ results in $y(k)$ output and $3u(k)$ results in $3y(k)$ output, is the system linear? (5 points)
9. If you have systems with the transfer functions below, are they BIBO stable? Why? (10 points)
 - a. $G(z) = 0.3/(z-0.3)$
 - b. $G(z) = 0.6/(z+1.1)$
10. Explain the derivation of the following formula that is used for computing settling time (10 points):

$$K \approx \frac{-4}{\ln |a|}$$

11. Suppose you want to control the energy and smart lights in a smart home. What is the target system, controller, and data (including the reference input, control input, and measured output)? Answers can be at a fairly high level. (10 points)

12. Find the steady state gain for the transfer function $F(z) = (-0.014KP)/(z - .6 - KP*0.011)$ when $KP = 1$ and when $KP = 100$. (10 points)